Lecture 10?

4.12.2025

$f: (B_2)^n \to B_2$ boolean function of n variables.

The support of f

Ex: $n=4 \Rightarrow m = x_1 \wedge \overline{x_3} \to S_m = \{(1,1,0,1),(1,0,0,1),(1,1,0,0),(1,0,0,0)\}$.

Def 3: 1) adjacent monoms (they $\neq$ only by the power of the variable with t_i).

2) factorization of the monoms.

$m \vee m'$ contains the common part of $m \vee m'$.

$n = 3$ var

$m_5 = x_1^{1} \wedge x_2^{0} \wedge x_3^{1} = x_1 \overline{x_2} x_3$ $\qquad 5 = (101)_2$

$m_4 = x_1^{1} \wedge x_2^{0} \wedge x_3^{0} = x_1 \overline{x_2}\, \overline{x_3}$ $\qquad 4 = (100)_2$

they $\neq$ by a single variable $(x_3, \overline{x_3})$

$\Rightarrow m_5 \vee m_4 = x_1 \overline{x_2} x_3 \vee x_1 \overline{x_2}\, \overline{x_3} = x_1 \overline{x_2} \underbrace{(x_3 \vee \overline{x_3})}_{1} = x_1 \overline{x_2}$.

This is simpler from a sintactic point of view (what m_5, m_4 have in common)

Concatenation = Conjunction.

Def: max monom for a function $(\Leftrightarrow)$ $\begin{cases} m \le f \\ m < m' \le f \end{cases}$

$= x_2x_3 = m_6 \vee m_2$ $\quad max_4 = x_2x_3 = m_3 \vee m_7$

$M(f) = \{ max_1, max_2, max_3, max_4\}$: **maximal monoms**

$C(f) = \{max_3, max_4\}$: **central monoms**

Second case of the *simplification algorithm.*

We denote by $g(x_1, x_2, x_3) = max_3 \vee max_4$

There are two **disjunctive simplified forms** of f:

$f_1^{DS}(x_1, x_2, x_3) = g \vee max_1 = max_3 \vee max_4 \vee max_1 =$

$= x_2x_3 \vee x_2x_3 \vee x_1x_2$

$f_2^{DS}(x_1, x_2, x_3) = g \vee max_2 = max_3 \vee max_4 \vee max_2 =$

$= x_2x_3 \vee x_2x_3 \vee x_1x_3.$

$DCF(f) = m_1 \vee m_2 \vee m_4 \vee m_5 \vee m_6$

$= x_1^0 x_2^0 x_3^1 \vee \ldots$

$CCF(f) = M_0 \wedge M_3 \wedge M_7$

$= (x_1^0 \vee x_2^0 \vee x_3^0) \wedge \ldots$

$= (x_1^1 \vee x_2^1 \vee x_3^1) \wedge \ldots$

DCF

Karnaugh diagram

$x_1 \backslash x_2x_3$	00	01	11	10
0		m_1		m_2
1	m_4	m_5		m_6